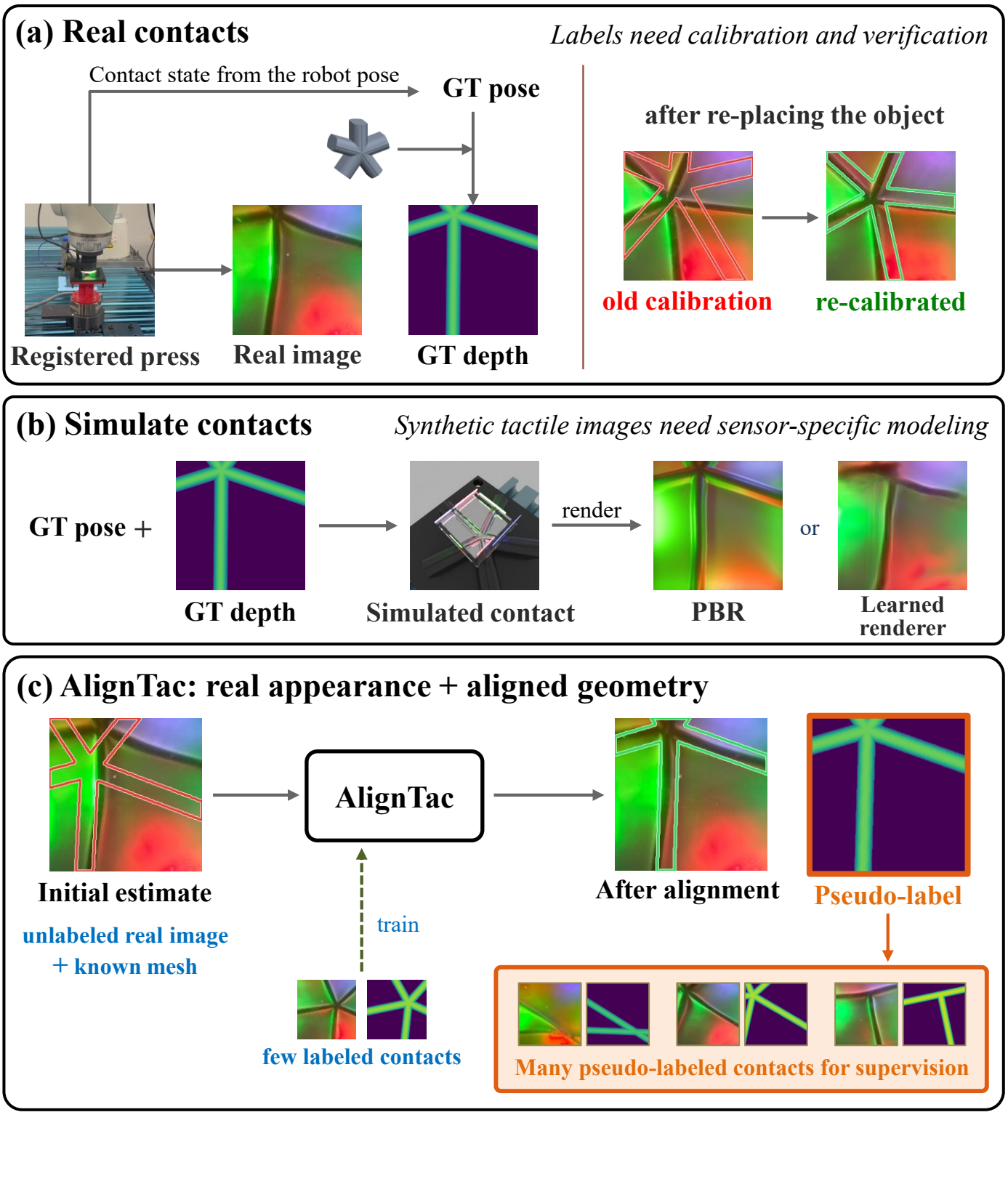

(a) Real contacts
Labels need calibration and verification
Contact state from the robot pose
GT pose
Registered press
Real image
GT depth
after re-placing the object
old calibration
re-calibrated
(b) Simulate contacts
Synthetic tactile images need sensor-specific modeling
GT pose +
GT depth
Simulated contact
render
PBR
or
Learned renderer
(c) AlignTac: real appearance + aligned geometry
Initial estimate
unlabeled real image + known mesh
AlignTac
train
few labeled contacts
After alignment
Pseudo-label
Many pseudo-labeled contacts for supervision